

TANVIR AHMED

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RESEARCH INTERESTS

Efficient multimodal inference and the scheduling of computation across vision-language models of different sizes; video understanding and its evaluation; calibration, abstention and uncertainty; explainable and auditable systems; empirical audits of quantum machine learning; and inference on microcontroller-class hardware.

EDUCATION

North South University Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering 2022 – Dec 2026 (*expected*)
• Department of Electrical and Computer Engineering. CGPA 3.02 / 4.00.
Milestone College Dhaka, Bangladesh
Higher Secondary Certificate, Science 2021
• GPA 5.00 / 5.00.

PUBLICATIONS

Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency ACCEPTED
Tanvir Ahmed. *LM-KBC 2026 shared task (5th edition), Automated Knowledge Base Construction (AKBC) at EMNLP 2026*.

- Set-valued knowledge base construction with an open-weight model of at most 32B parameters, no retrieval and no fine-tuning. Because a correct empty prediction scores a perfect F_1 , the system treats *when to answer* as a first-class decision.
- Scores 0.461 relation-averaged and 0.488 micro macro- F_1 on validation against 0.300 / 0.313 for the official baseline, beating it on five of the six relations. Code: github.com/minhaz-42/lm-kbc-2026.

SoundSentry: A Low-Cost Edge-AI Acoustic Event Detection and Direction-Finding System on STM32 UNDER REVIEW
Tanvir Ahmed, Sanzana Afrin Tonny, Husain Kabir Sifat. *International Conference on Electrical and Computer Engineering (ICECE) 2026, Dhaka*.

- STM32F103 prototype with a three-microphone array, ADC-DMA acquisition, on-device classification over 11 features, servo direction response and SD-card logging.
- 59.55% accuracy and 58.17% macro- F_1 on the 2,000-clip ESC-50 dataset under five-fold validation.

RESEARCH

Frame-Aware Multi-Model Scheduling for Efficient Video Captioning Senior Design Project, 2026
• Routes video captioning between a 4B and a 27B vision-language model using a CLIP-plus-graph-attention complexity scorer trained without human labels.
• Answers a 19-clip benchmark in 70 API requests against 228 for the small model alone, scoring above it on every judged axis. Lead author of four.

Quantum-Enhanced Driver Distraction Detection Directed Research, 2026
• Hybrid classical-quantum pipeline with an entropy gate and a continuous risk score, under a protocol where only the classification head differs.
• The quantum head reaches accuracy parity with its matched classical control and cuts expected calibration error by 37% relative; the report states plainly that the gain cannot be attributed to the circuit.

Observing Is Not Free: the decision value of visual observations in vision-language agents In preparation
• Measures whether an agent's observation-selection criterion ranks observations the way their realised effect on the answer does. On V*Bench, 8.2% of observations help and 5.9% actively harm.

Knowing What You No Longer Know: calibrated belief validity for persistent world memory In preparation
• Separates whether a belief is still true from how confident the model was when it formed it, and builds a benchmark of elapsed-time-matched pairs to score it.

SimCert: a classical-simulability audit of trained QNLP text models Abstract submitted, QTML 2026
• Re-simulates trained quantum-NLP circuits under matrix-product-state truncation to test whether the quantum component is load-bearing.

Sequential Video Belief In progress

- Whether a video-language model's own stated answer impairs its ability to revise that answer when later evidence contradicts it.

Uncertainty-Guided Language-Grounded 3D Scene Graphs from Sparse Views

In progress

- Frozen geometry and vision-language models with a lightweight graph reasoning module, for 3D object segmentation and spatial relations from a few casual images.

TECHNICAL SKILLS

Programming: Python, C++, C, Java, JavaScript, TypeScript, PHP, SQL, L^AT_EX

Machine Learning: PyTorch, Transformers, PyTorch Geometric, scikit-learn, graph attention networks, fine-tuning, calibration, Hydra

Vision & Multimodal: OpenCV, CLIP, DINOv2, SAM 2, MediaPipe, vision-language models, video understanding, Ollama

Quantum ML: PennyLane, variational circuits, OpenQASM, matrix-product-state simulation, lambeq

Web & Software: Django, React, Node.js, REST APIs, WebSocket, Docker, PostgreSQL, MySQL, SQLite

Embedded: STM32, embedded C, ADC-DMA acquisition, sensor interfacing

Graphics & Games: Godot 4, Blender, Metal, SDL3, CMake

Tools: Git, Linux, Jupyter, Apple MPS, VS Code

PROJECTS

Pakna Council Local-first web app running several LLMs in parallel on one prompt through Ollama. Django, React, Docker, WebSocket. Source · demo.

SHAND Deterministic, explainable assumption detector for English text; every flag traces to its rule. Python, Django, D3.js. Source · demo.

ISKA Privacy-first document analysis assistant with a Chrome extension. TypeScript, Chrome extension APIs. Source · demo.

Nestledown Cozy 3D exploration game built solo in Godot 4 with every asset script-generated; a parallel build runs on a custom C++20 engine with a Metal renderer. Trailer.

Entangled Fusion Quantum Circuits Multimodal platform pairing vision analysis with quantum-inspired fusion and language reasoning. Python, Django, PennyLane, OpenCV.

SUMBA 3D sign language translation: gesture capture, dataset building and model training. Django, MediaPipe, Transformers, React Native.

FitWell Nutrition and health assistant with meal planning and BMI tracking. Django, Python, Qwen, SQLite.

DG Healthcare Management System Records, scheduling, prescriptions and billing with role-based access. PHP, MySQL, JavaScript.

ACTIVITIES

NSU Trimodal AI Lab, Research Member

2025 – present

- Collaborative experimentation on multimodal and agentic AI systems; idea validation, prototyping and literature review.

IEEE NSU Student Branch, General Member

2023 – present

- Technical workshops, seminars and community outreach programmes.

REFERENCE

Dr. Md Adnan Arefeen

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Contact details available on request.